

Skills Summary

Using u-Substitution in Indefinite and Definite Integrals

Use this reference while completing your worksheet or when studying.

The chain rule, read backwards. If $u = g(x)$ then $du = g'(x) dx$, and $\int f(g(x)) g'(x) dx = \int f(u) du$.

- **Choose** u to be the inner function — what is under a root, inside a bracket raised to a power, in an exponent, or in a denominator.
- **Test it:** write du and check that the rest of the integrand is a *constant* multiple of it. A leftover x means the choice was wrong.
- **Indefinite:** back-substitute and add $+C$. **Definite:** convert both limits to u and never return to x .

1. When the derivative is already there

The easiest case: the integrand is exactly $f(u) du$. Name u , write du , swap everything, integrate with the power rule, back-substitute.

Example: Find $\int 3x^2 (x^3 + 2)^4 dx$.

Step 1. The bracket is the inner function, and its derivative $3x^2$ is already outside, so no adjusting factor is needed.

Step 2. With $u = x^3 + 2$ and $du = 3x^2 dx$, the integral is $\int u^4 du = \frac{u^5}{5}$.

$$\frac{1}{5} (x^3 + 2)^5 + C$$

Try it: Find $\int 2x (x^2 - 7)^5 dx$.

Check: $\frac{d}{dx} \left(\frac{2}{6} (x^2 - 7)^6 \right) = 2x (x^2 - 7)^5$

2. When a constant factor is missing

If $du = k g'(x) dx$ but the integrand supplies only $g'(x) dx$, multiply the integral by $\frac{1}{k}$. Constants move across the integral sign freely; variables never do. If you have to move an x , change your choice of u .

Example: Find $\int x \sin(x^2) dx$.

Step 1. Take $u = x^2$ and compare $du = 2x dx$ with the $x dx$ actually present — the gap is a constant, so it is fixable.

Step 2. $x dx = \frac{1}{2} du$, so the integral is $\frac{1}{2} \int \sin u du = -\frac{1}{2} \cos u$.

$$-\frac{1}{2} \cos(x^2) + C$$

Try it: Find $\int x^2 e^{x^3} dx$.
 Check: $\frac{1}{3} e^{x^3} + C$

3. Definite integrals: change the limits

Convert both bounds with the same rule $u = g(x)$, then evaluate entirely in u . This is faster than back-substituting and it removes the classic error of evaluating a u -antiderivative at the old x -limits.

Example: Evaluate $\int_0^1 2x (x^2 + 1)^3 dx$.

Step 1. Convert the limits along with the integrand so the answer never has to return to x .

Step 2. $u = x^2 + 1$ gives $du = 2x dx$, with u running from 1 to 2, so the integral is $\int_1^2 u^3 du = \left[\frac{u^4}{4} \right]_1^2 = \frac{16-1}{4} = \frac{15}{4}$.

Try it: Evaluate $\int_0^1 3x^2 (x^3 + 1)^2 dx$.
 Check: $\frac{1}{3} (x^3 + 1)^3 + C$

4. Improper integrals: substitute, then take a limit

$\int_a^\infty f = \lim_{b \rightarrow \infty} \int_a^b f$. Do the substitution on the proper integral over $[a, b]$, evaluate, and take the limit last. A finite limit means the integral converges to it; an infinite limit means it diverges.

Example: $\int_0^b \frac{2x}{(x^2 + 1)^2} dx$ equals $1 - \frac{1}{b^2 + 1}$. What happens as $b \rightarrow \infty$?

Step 1. Take the limit of the evaluated antiderivative, since the substitution work is already finished.

Step 2. As b grows, $\frac{1}{b^2 + 1} \rightarrow 0$, so only the constant term survives and the integral converges.

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Try it: Evaluate $\int_0^\infty \frac{4x}{(x^2 + 1)^2} dx$.
 Check: $-\frac{2}{x^2 + 1} + C$